1 Introduction

1.1 Holomorphic functions

Definition 1.1: Holomorfic function

A function $f\colon \mathbb{C} \supseteq \Omega \to \mathbb{C}$ is *holomorphic* if it is complex differentiable on Ω, meaning that for any $z \in \Omega$

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$$

exists. Equivalently,

$$f(z+h) = f(z) + hf'(z) + o(h).$$

Equivalently, f is differentiable as a function $\mathbb{R} \to \mathbb{R}$ and its differential is a matrix of multiplication by a complex number

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

for some reals a, b. Writing this using partials, we get

Definition 1.2: Cauchy - Riemann equations

Let f be holomorphic and write $f(x+iy) = u(x,y) + iv(x,y)$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

TODO: $\frac{\partial}{\partial z}$, $\frac{\partial}{\partial \bar{z}}$.

Lemma 1.3: Closedness of fdz

Let $dz = dx + idy$. Then f is holomorphic iff the form fdz is closed, meaning

$$d(fdz) = 0.$$

All arithmetic properties are the same as for reals.

Example 1.4: Some holomorphic functions

Any polynomial, convergent power series (not proven yet), exp, all trigonometric functions outside their poles. sin and cos are unbounded for large y and tan is bounded for large y.